BEING (LOCALLY) QUASI-FINITE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one same-chapter lemma, one Morphisms of Spaces lemma and one Descent lemma plus the framework lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- fpqc descent of quasi-compactness. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-quasi-compact. Supplies the reduction from quasi-finite to locally quasi-finite.
- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure that the proof verifies.
- Local quasi-finiteness is \'etale-local on the source. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-finite-local. Supplies assumptions (1), (2) and the chart reductions.
- fpqc descent of local quasi-finiteness for schemes. Stacks Project, Descent, Lemma descent-lemma-descending-property-quasi-finite. Supplies the final descent step for U \to Z.

Source reference. Original lemma 041L: sources/stacks-chow/tags.txt, source lines 5296–5296. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 040Z: sources/stacks-chow/tags.txt, source lines 5274–5274. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02VI: sources/stacks-chow/tags.txt, source lines 3769–3769. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The properties $\mathcal{P}(f) =$"f is locally quasi-finite" and $\mathcal{P}(f) =$"f is quasi-finite" are fpqc local on the base.*

Proof. We have already seen that "quasi-compact" is fpqc local on the base, see Lemma 041L. Hence it is enough to prove the lemma for "locally quasi-finite". We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemma 040Z. Let $Z' \to Z$ be a surjective flat morphism of affine schemes over S. Let $f : X \to Z$ be a morphism of algebraic

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spaces, and assume that the base change $f' : Z' \times_Z X \to Z'$ is locally quasi-finite. We have to show that f is locally quasi-finite. Let U be a scheme and let $U \to X$ be surjective and étale. By Morphisms of Spaces, Lemma 040Z again, it is enough to show that $U \to Z$ is locally quasi-finite. Since f' is locally quasi-finite, and since $Z' \times_Z U$ is a scheme étale over $Z' \times_Z X$ we conclude (by the same lemma again) that $Z' \times_Z U \to Z'$ is locally quasi-finite. As $\{Z' \to Z\}$ is an fpqc covering we conclude that $U \to Z$ is locally quasi-finite by Descent, Lemma 02VI as desired. $\square$